

Let (II) be true for all $n \leq k$, and consider for $n=k+1$

$$F_{k+1} = F_k + F_{k-1} > k + (k-1) > k+6 > k+1, \text{ for all } k > 7$$

To show F_n is not bounded above, consider positive M , as large as we like. Then $F_M > M$. Hence the sequence is not bounded above.

Q5. Show that the sum of first n terms of Fibonacci sequence $\sum_{i=1}^n F_i = F_{n+2} - 1$. For all n .

Ans.

Proof: We prove the claim by induction on n .

Base case: For $n=1$, $\sum_{i=1}^1 F_i = F_1 = 0$. Also, $F_{1+2} - 1 = F_3 - 1 = 1 - 1 = 0$

Induction Hypothesis: Let the claim be true for

$$n = k, \text{ i. e., } \sum_{i=1}^k F_i = F_{k+2} - 1 \quad \dots \dots \dots (i)$$

Induction Step: Consider $\sum_{i=1}^{k+1} F_i = \sum_{i=1}^k F_i + F_{k+1} = (F_{k+2} - 1) + F_{k+1}$ (by using (i) above)

$$= F_{k+2} + F_{k+1} - 1 = F_{k+3} - 1.$$

Hence the proof.

Q6. Show that every finite set $\{a_1, a_2, \dots, a_n\}$ is bounded.

Ans.

Let $m = \min. \{a_1, a_2, \dots, a_n\}$, and $M = \max. \{a_1, a_2, \dots, a_n\}$. Then $m \leq a_i \leq M$, for all i .

Q7. Show that union of closed intervals is a closed set.

Ans. We prove the result for union of two closed intervals. The same proof procedure can be applied to union of finite number of closed intervals. Let $A = [a_1, b_1] \cup [a_2, b_2]$. Assuming $b_1 < a_2$. Then consider $R - A = (-\infty, +\infty) - ([a_1, b_1] \cup [a_2, b_2]) = (-\infty, a_1) \cup (b_1, a_2) \cup (b_2, +\infty)$, where each set on R.H.S. is an open interval. Hence, $R - A$ is an open set, and A is a closed set.

Q8. Show that an open interval is not a closed set

Ans. For the open interval (a, b) , its complement $R - (a, b) = (-\infty, a] \cup [b, +\infty)$ is union of semi-open intervals, and hence is not open.

Q9. Show that a semi-closed interval is not a closed set.

Ans. For a semi-closed interval $(a, b]$, its complement $R - (a, b] = (-\infty, a] \cup (b, +\infty)$. But $(-\infty, a]$ is not an open set. Hence, semi-closed interval $(a, b]$ is not closed.

UNIT 14 CALCULUS OF SEVERAL VARIABLES

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14.0 OBJECTIVES

After going through this unit, you will be able to

- Explain & define the concept of *limit*, and compute it for functions of two or more variables;
 - Explain & define the concept of *partial derivative*;
 - Compute *partial derivatives* for function of two or more variables, when function is given explicitly, implicitly, or as functional composition;
 - Define & compute directional derivatives;
 - Explain & define the concept of function from R^n to R^m ;
 - Explain the terms *Jacobian Matrix*, *Hessians*, & compute these;
 - State Mean Value Theorem, explain its significance, and use it in solving problems involving one, two or more variables;
 - Explain Taylor's approach to evaluation of functions through polynomial approximation;
 - State Taylor's theorem, explain its significance, & use it in polynomial approximation; and
 - Explain and derive Taylor's series for functions having derivatives of all orders.
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14.1 INTRODUCTION

The topic of '*Calculus of Several Variables*' is also known, alternatively, as '*Multivariable calculus*' or '*Multivariate calculus*.' It is an *extension of the calculus of one variable*—discussed in the previous block—to *calculus with functions of several variables*. There is possibility of some overlap of the contents of this unit with those of Unit 10 & Unit 11 of the previous block. Also, in view of the fact that Real Analysis and Calculus are two different, but inseparable, facets of the same part of mathematics, viz. *continuous Mathematics*, which is concerned with *limiting processes* and which involves concepts like limit, continuity, convergence, differentiability etc., **again**, there is possibility of overlap with Unit 13—which deals with Real Analysis of several variables—of this block. Care has been taken to reduce the overlap, in both the cases, to the bare minimum.

We know that the concept of '**limit**' underlies all other basic concepts—of continuous Mathematics—including continuity, convergence, differentiation & differentiability, integration and integrability etc. *Just to confirm the preceding statement, please go, once again, through Block3: Calculus, and Unit13: Real Analysis of this block.* You would not find a single basic concept which does not involve limit in its definition. *However, almost whole (except a small final part of Unit 13) of the discussion is **only** in respect of functions of **single** variable.*

Extension of the concept of limit—and hence, of each of the other concepts like differentiability and integrability—from functions of **one** variable to functions of

two variables, is surprisingly, ***much more complicated than*** the extension of these concepts from functions of *two* variables to ***any arbitrary*** finite ***number*** (say 100) of variables. In view of the facts of the foundational role of limit in continuous Mathematics & the complexity involved in extending it, we first discuss ‘limit’ in the next section.

14.2 THE CONCEPT OF ‘LIMIT’

Let us recall the following two examples about the limit of real-valued real functions:

Case (i): $\lim_{x \rightarrow 0} (1/x^2)$, **exists** and equals $+\infty$.

However,

Case (ii): $\lim_{x \rightarrow 0} (1/x)$, **does not exist**,

As the *right-hand* limit $\lim_{x \rightarrow 0^+} (1/x) = +\infty$ **and** the *left-hand* limit $\lim_{x \rightarrow 0^-} (1/x) = -\infty$ *do not agree*.

We summarize the reason for the existence of limit in case (i) above, and non-existence of limit in case (ii), from the broader perspective of generalisation to functions of two or more variables.

For this purpose, *first of all, we briefly explain why, at all, the concept of limit is defined and used*: The concept is used in view of the fact that a function may *NOT* be defined at some point, e. g. the function

$f: R - \{5\} \rightarrow R$ with $f(x) = (x^2 - 25) / (x - 5) \dots \dots$ **(a)**

is **not** defined at $x = 5$, because the denominator $(x - 5)$ becomes 0 at $x = 5$. However, $f(x) = (x^2 - 25) / (x - 5) = (x+5)$ for $x \neq 0 \dots \dots$ **(a1)**

The idea of limit is to assign SOME value to the functions like the function f under (a) above, at the point(s), e. g. $x=5$, in our case, where it is not defined. But we do not do the assignment of intended value arbitrarily. The idea of ‘limit’ is to assign a value, which ***is consistent with*** the values ***at ALL*** of its neighbouring points:

For example, in the case of (a) above, ***under the idea of limit***, we ***attempt*** to assign a possible value to $f(x)$ at $x=5$, so that the ***intended-to-be-assigned value***

i) ***differs by a small amount from the (actual) values***, at neighbouring points $(5 + \varepsilon)$ and $(5 - \varepsilon)$, ***at which*** using (a1) above, we get the values

$$f(5 + \varepsilon) = (5 + \varepsilon + 5) = 10 + \varepsilon,$$

and

$$f(5 - \varepsilon) = 10 - \varepsilon,$$

where $\varepsilon > 0$ is taken as very small number: as small as we like, but positive.

($\varepsilon > 0$, generally taken as a very small number, is used for defining a neighbor at a distance ε)

- ii) the actual values **differ from each other** by a small amount, and
- iii) as we get closer to 5 (in our case, by taking smaller and smaller values of ε), the actual values also become still closer to each other and also to the intended value.

The attempt may succeed, or may *not* succeed as discussed below for *case (i)* and *case (ii)*, with which we started our discussion:

In Case-i: $\lim_{x \rightarrow 0} (1/x^2)$, the neighbours of 0 are $(0 + \varepsilon) = (\varepsilon)$, and $(0 - \varepsilon) = (-\varepsilon)$, where $\varepsilon > 0$ is initially taken small, and gradually made still smaller. In the process, the values $[1/(\varepsilon)^2]$ and $[1/(-\varepsilon)^2]$ of $(1/x^2)$ become larger, and both positive in sign.

For example, for $\varepsilon = 1/1000 (= 10^{-3})$, both values become $1000000 (= 10^6)$. Next, we take still closer neighbours of $x=0$ by taking, for example, $\varepsilon = 1/100000 = 1/(10)^5$, then both values become $100000000000 = (10)^{10}$, which is still bigger a number. Thus, as the independent variable x approaches 0, from **two different directions, from positive side, and from negative side** of 0, the corresponding values are equal and become larger and larger moving toward ∞ .

Thus, the intended value $+\infty$ satisfies all the three conditions (i), (ii) and (iii) above. Hence, the limit exists and equals to $+\infty$.

In Case-ii: $\lim_{x \rightarrow 0} (1/x)$, again the neighbours of $x=0$ are $(0 + \varepsilon)$ and $(0 - \varepsilon)$, where $\varepsilon > 0$ is initially taken small, and gradually made still smaller. In the process, values $[1/(\varepsilon)]$ and $[1/(-\varepsilon)]$ become larger in magnitude, **but opposite in sign**. For example, $\varepsilon = 1/1000 (= 10^{-3})$, the values become respectively $+(10^3)$ and $-(10^3)$. **The two values are far apart.**

Next, we take still closer neighbours of $x=0$, by taking for example, $\varepsilon = 1/100000 (= 1/(10)^5)$, then values become respectively $+100000$ and -100000 , **which are still farther apart.**

Thus, as we approach 0, from two different directions, from the positive side, and from the negative side of 0, the corresponding values are drifting farther and farther apart.

Values at **neighbouring points from different directions** drastically **differ**. Hence, **$\lim_{x \rightarrow 0} (1/x)$ does not exist.**

But most of the above has already been discussed in one form or another in Block 3: Calculus, specially, in Unit 9: Functions, Limit and Continuity.

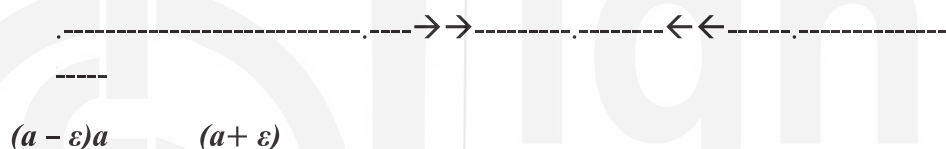
So, from the perspective of the extension of the concept of 'limit of functions of one variable' to 'limit of functions of two or more variables', what is new?

For this purpose,

- i) let us recall ‘*The idea of limit is to assign SOME value to the function— at a given point where it is not defined—but not arbitrarily. The idea of ‘limit’ is to assign a value, which is **consistent with** the values **at ALL** of its neighbouring points*’, and
- ii) note the phrase ‘**at ALL**’ in the phrase ‘*consistent with the values **at ALL** of its neighbouring points*’, and
- iii) for finding ALL possible neighbours of a point, let us **visually examine**
 - a) the real line, representative of $\mathbb{R}$, the domain of functions of *single* variables, and
 - b) two-dimensional plane, representative of $\mathbb{R}^2$, the domain of functions of *two* variables,

as follows

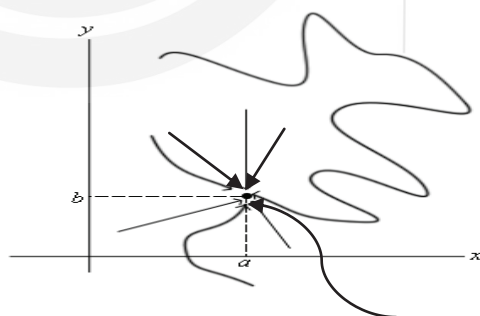
- i) ON REAL LINE:



The neighbours of point a might approach the point a from only two directions:
(i) from LHS (i.e. $(a - \varepsilon) \rightarrow a$) and (ii) from RHS (i.e. $(a + \varepsilon) \rightarrow a$), for $\varepsilon > 0$.

But

- ii) In two-dimensional plane: The point (a, b) may be approached, by its neighbours from infinitely many directions, as shown below:



(Source: Paul Dawkins)

Thus, the problem of existence of limit at a point (a, b) depends on conciliation of values at neighbouring points on paths along infinitely many directions.

The above explanation gives a glimpse in the **difficulty & and possible impediments and pitfalls** in the process of extending the concept of limit of functions of **one** variable to limit of functions of **two** variables.

This also gives an idea of the difficulties which may be encountered in extending the other basic concepts—from **one**-variable functions to **two**-variable

functions—of calculus, because each is defined in terms of limit, viz. the *derivative* at a point, say, $x = a$, if it exists, is given by

$$f'(x)_{x=a} = \lim_{h \rightarrow 0} [f(a+h) - f(a)]/h.$$

Thus, derivative, at basic level, is a limit of some function.

In view of the above explanation, in the process of extending the concept of limit from functions of one variable to functions of two variables, we not only have the concepts of various *partial & higher order derivatives*, but also *directional derivatives* etc.

Remarks: Extension of the concepts of limit, differentiability etc. of functions of *two variables* to *n variables*—where $n > 2$, may be any integer, say even 50 or 100—is almost straight forward. This is in view of the fact that any point $(a_1, a_2, \dots, a_n)$, in the case of n variables, may be approached by its neighbours through possibly infinitely many directions, as is the case of functions of *two* variables.

14.3 COMPUTING LIMIT FOR FUNCTIONS OF TWO OR MORE VARIABLES

The *derivatives and partial derivatives of a function* of two or more variables, at any point in its domain of definition, *are ultimately limits* of some related function.

For example, from Unit 11, we know that for the function $U = f(x_1, x_2)$, of two independent variables x_1 , and x_2 , the partial derivative of U with respect to x_1 , denoted by $\frac{\partial U}{\partial x_1}$, $\frac{\partial f}{\partial x_1}$ or simply f_1 is defined as

$$f_1 = \lim_{\Delta x_1 \rightarrow 0} \frac{\Delta U}{\Delta x_1} = \lim_{\Delta x_1 \rightarrow 0} \frac{f(x_1 + \Delta x_1, x_2^0) - f(x_1, x_2^0)}{\Delta x_1},$$

and, similarly, with respect to x_2 , partial derivative f_2 is defined as

$$f_2 = \lim_{\Delta x_2 \rightarrow 0} \frac{\Delta U}{\Delta x_2} = \lim_{\Delta x_2 \rightarrow 0} \frac{f(x_1^0, x_2 + \Delta x_2) - f(x_1^0, x_2)}{\Delta x_2}.$$

We may note that each of $\lim_{\Delta x_1 \rightarrow 0} f$ and $\lim_{\Delta x_2 \rightarrow 0} f$ is actually limit of only one variable, as the other variable is kept constant.

However specially, in view of the discussion in the previous section, *the task of computing limit*—of, say, function $U = f(x_1, x_2)$ of two variables at a point (a, b) of domain of its definition—is *quite a different, and probably quite daunting, task as, for the limit of a function of two variables to exist*, the values at neighbouring points of (a, b) from *potentially infinite directions* must reconcile.

Also, except in these definitions, we did not discuss the topic of how to calculate limit of a function of two or more variables. Again, in view of the significance of this type of limit as fundamental concept, we need to discuss the topic in some

detail. Before we enter into final details, let us know that *either of the following notations is used to denote limit of a function $f(x, y)$ at point (a, b) of its domain of definition:*

$$\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) \quad \text{or} \quad \lim_{(x,y) \rightarrow (a,b)} f(x, y)$$

Generally, the latter one is preferred.

In this section, through some examples, we explain the process of finding limits of such functions.

Again, in view of the above discussion, the task seems to be challenging. However, the task is facilitated by the following relevant facts

- i) For a function—whether of one variable or two variables—to be *continuous* at a point, by definition, the **VALUE** at the point should equal the **LIMIT** at the point. Thus, for a function $f(x, y)$ of two variables, to be *continuous* at point (a, b) , we need to have

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

- ii) Graphically, a function—of one variable (or, of two variables)—is continuous at a point $x=a$ (or, at $(x, y) = (a, b)$), if it does not have a hole in the neighbourhood of the point, on the curve (or, in the surface).
- iii) *If* any function of **two variables**—whether it is a polynomial, trigonometric function, exponential or logarithmic function, or any combination of these — **is continuous when** it is made a function of **one variable** by taking $x=y$, or by treating one of the variables as a constant, **then** it is also continuous as a function of two variables.

And, in the case of *continuous* function, the required **LIMIT** at a point equals the **VALUE** at that point. Hence, for finding the required **limit**, we just calculate its **value**.

Example 14.1 Calculate

$$\lim_{(x,y) \rightarrow (1,3)} \sin((\pi/2) x) + 3 y^2 + 7,$$

First, we note that the function $\sin((\pi/2) x) + 3 y^2 + 7$ is continuous everywhere, and hence, at $(1, 3)$ also. Therefore, for calculating the limit, we *will not try* to approach its value from potentially infinite directions, but *just find its value at $(1, 3)$* , and take this value as the value of the required limit. Thus,

$$\lim_{(x,y) \rightarrow (1,3)} \sin((\pi/2) x) + 3 y^2 + 7 = \sin((\pi/2) \cdot 1) + 3(3)^2 + 7 = 1 + 27 + 7 = 35.$$

- iv) While using (iii) above, only we have to be careful about **possible problematic conditions**: (i) divisions by zero, (ii) square-rooting negative numbers, and (iii) taking logarithms of zero, or of negative numbers.

- v) The possible problematic conditions of (iv) above, ***may be only apparently present, and may be removed.*** Then again, the limit may exist and be calculated.

Example 14.2 Suppose we intend to discuss the following limit

$$\lim_{(x,y) \rightarrow (1,1)} (x^2 - 2xy + y^2) / (x^2 - y^2),$$

The function is not continuous at point (1, 1), as at this point, the denominator becomes zero.

However, except for the line $x = y$, at which the point (1, 1) lies, and at which the limit is intended to be calculated, the function $(x^2 - 2xy + y^2) / (x^2 - y^2)$ equals the function $(x - y) / (x + y)$. And, we know, in calculating the limit at a point (here $(x, y) = (1, 1)$), the value at the point is NOT taken into consideration. Therefore, for the purpose of limit

$$\lim_{(x,y) \rightarrow (1,1)} [(x^2 - 2xy + y^2) / (x^2 - y^2)] = \lim_{(x,y) \rightarrow (1,1)} [(x - y) / (x + y)] = 0/2 = 0.$$

The limit has been calculated by removing the problematic condition, i. e. *division by zero*, at the point/line where limit is to be computed.

- vi) *If, the problematic condition cannot be removed, then there is chance that the limit **does not** exist.*

Now, whether the problematic condition cannot be removed, or for any other reasons, if the limit of a function of two variables **does not exist** at a point say (a, b), then we **need not consider** values at neighbouring points **along all the infinite directions**, but **along only those** two **directions** which propose **different values** for the intended limit.

Example 14.3

Determine if the following limit exists or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (0,0)} xy / (3x^2 + y^2)$$

Solution: Clearly the function is not continuous at the point (0, 0), as the denominator of the function is zero at (0, 0). Also, there is no common factor in numerator and denominator to cancel out. Hence, *there is chance that the limit does not exist.*

In such cases, we attempt to approach the point along two different curves along which possibility of disagreement in values of limits at (0, 0), in our case, is more. Generally, first we explore $x=0$, $y=0$, and $x=y$ etc.

We approach $(0, y) \rightarrow (0, 0)$, we get

$$\lim_{(0,y) \rightarrow (0,0)} xy / (3x^2 + y^2) = \lim_{(0,y) \rightarrow (0,0)} 0 \cdot y / (0 + y^2) = \lim_{(0,y) \rightarrow (0,0)} 0 / y^2 = 0$$

... (14.1)

Similarly,

$$\lim_{(x,0) \rightarrow (0,0)} x \cdot 0 / (3x^2 + 0^2) = \lim_{(x,0) \rightarrow (0,0)} 0 / (3x^2) = 0. \quad \dots (14.2)$$

However, this agreement in values of limits along two different paths **does not necessarily imply** existence of limit of the given function at the point. Because, there may be other paths along which disagreement in values of limits may lead to the conclusion of non-existence of limit. **There is need for further exploration.**

Another popular path for exploration in such cases is $x=y$. Let us explore it.

$$\lim_{(x,y) \rightarrow (0,0)} xy / (3x^2 + y^2) =$$

$$\lim_{(x,y) \rightarrow (0,0)} x \cdot x / (3x^2 + x^2) = \lim_{(x,y) \rightarrow (0,0)} 1/4 = 1/4. \quad \dots (14.3)$$

As (i) and (iii) do not agree, the required limit does not exist.

Check Your Progress 1

- 1) Determine if the following limit exist or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (4,1)} 3xy^2 / (x+y)$$

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- 2) Determine if the following limit exist or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (1, -1)} (x^2 + 2xy + y^2) / (x^3 + y^3)$$

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- 3) Determine if the following limit exist or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (1,1)} [2(1-y) + 3(x-1)] / (x-y)$$

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